# OpenWebSearch.eu — Large Scale Data for Statistical Scraping

Prof. Dr. Michael Granitzer Chair of Data Science, University of Passau Coordinator OpenWebSearch.eu

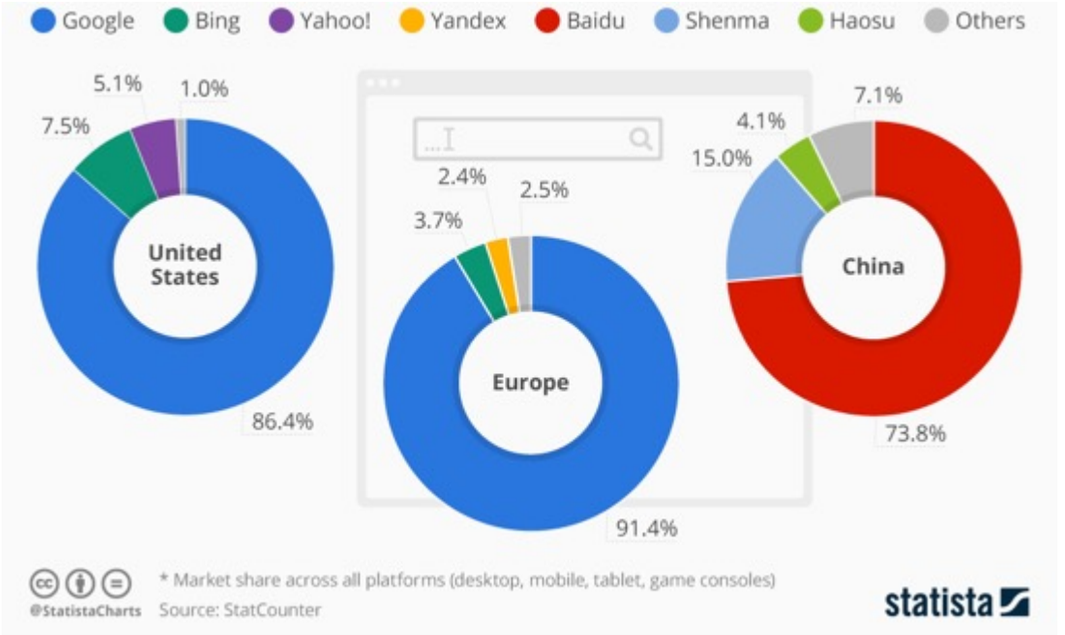


# Web Search Today

A critical resource managed as an oligopoly

# Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

> Being able to navigate the Web is as fundamental as GPS or the power grid

# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

**1.15 Billion**
Netcraft site responses
272.6 M domains · Dec 2024 Netcraft survey

**Top 0.1%–1% of sites**
may account for roughly half of all pages
heavy-tail intuition from large web graph studies

**~ 400 Billion**
documents in Google's search index
2020 figure cited in US v. Google testimony

*Sites ≠ domains ≠ pages; use exact counts cautiously*

### VELOCITY

**+8.6 M / month**
monthly net change in Netcraft site responses
Dec 2024 versus Nov 2024 Netcraft survey

**~ 3.2 / second**
same monthly net change, averaged across December

**149 ZB (global)**
all digital data created / copied worldwide in 2024
IDC / Statista global datasphere estimate

*The web changes constantly; crawling is continuous, not one-off*

### VARIETY

**150+ languages**
observed across websites
W3Techs usage survey

**249 CMS Platforms**
CMS platforms identified in the HTTP Archive dataset
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

**HTML · PDF · JSON-LD · media · APIs**
content types to parse and process

*HTML · PDFs · JS apps · schemas · media · APIs*

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista

- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

# Challenges

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- High infrastructure costs
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  - Web Technologies
- Growing legal uncertainties

High upfront costs to tap the Web as resources — especially small innovators and researchers are left behind

# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

1. **Open Technology Stack** Open-source pipelines for crawling, preprocessing, indexing
2. **Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
3. **Added Value Services** Dashboard, tools, data access APIs
4. **Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners



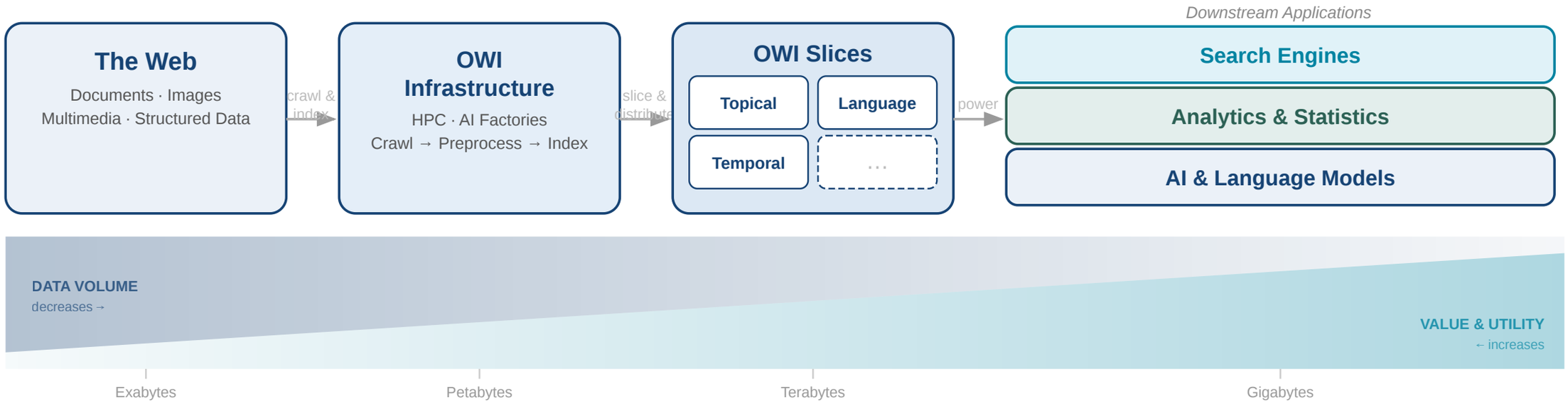

Plus 16 financially supported third-party projects (FSTPss)

Funded by **Horizon Europe** (NGI programme)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe




*Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." Journal of the Association for Information Science and Technology (2023).*

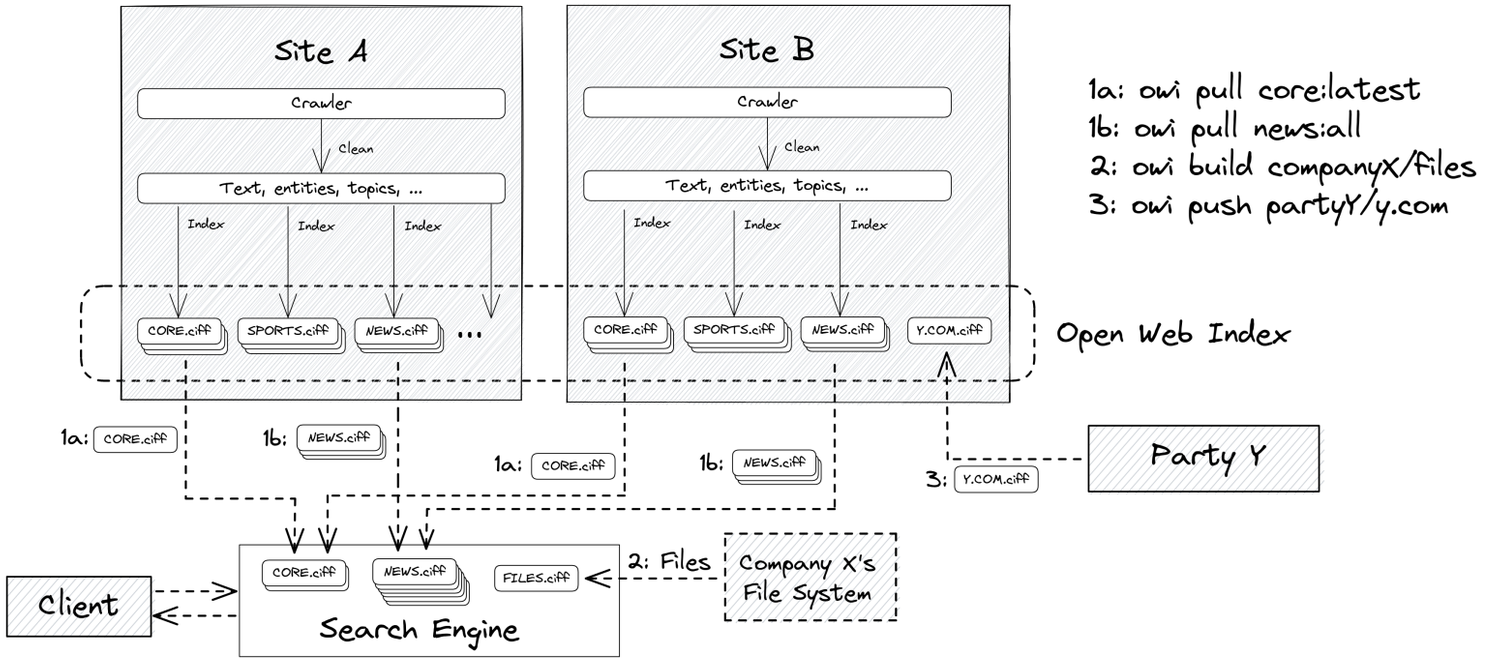
# What is the Open Web Index?

# The Open Web Index

> **The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.

# The Open Web Index

**The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.




### Federated production

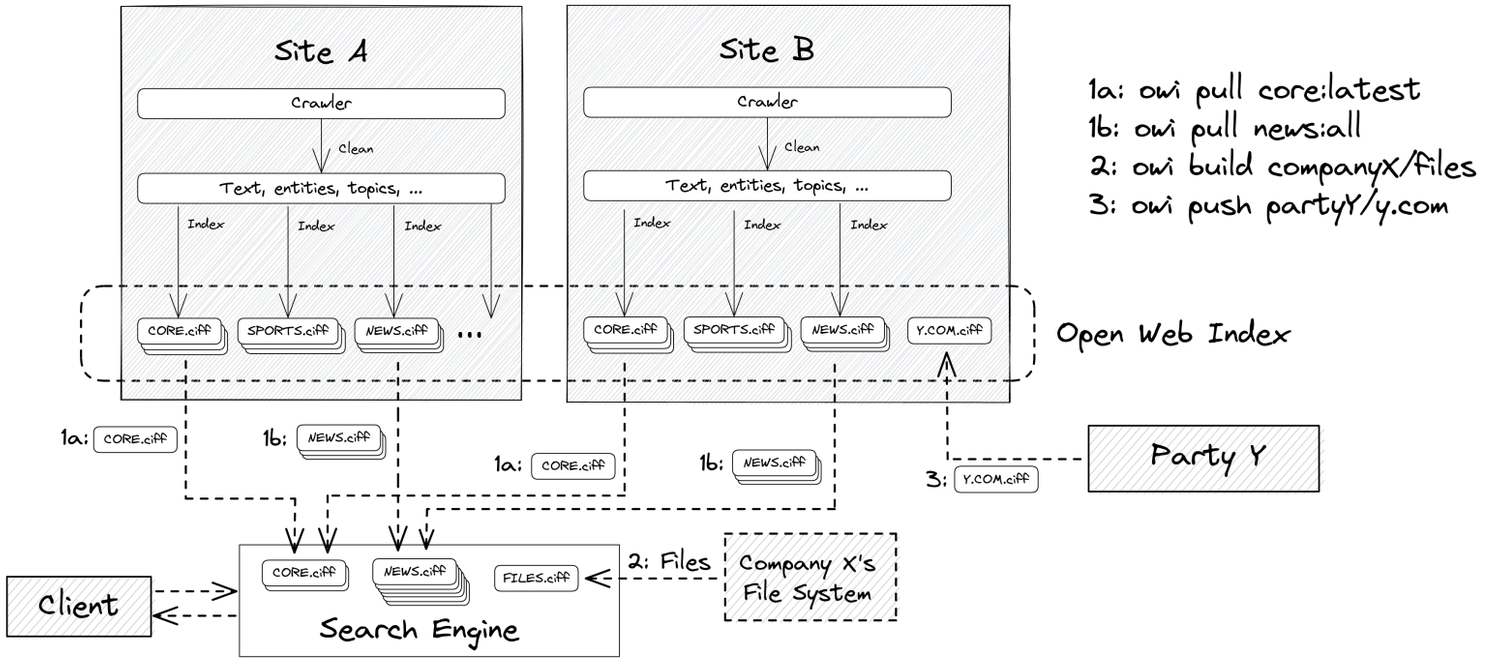
- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

### Three access verbs

- `owi pull` — download pre-built shards
- `owi build` — index your own data
- `owi push` — contribute shards back

# The Open Web Index

**The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.




### Federated production

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### Three output types per daily run — the building blocks of a shard:

- `index.ciff.gz` — sparse inverted index (CIFF format; pre-configured tokenizer)
- `metadata_*.parquet` — document metadata & enrichments (topics, microdata, locations, language, etc.; extensible)
- Dense embeddings (GPU-enriched slice, Jina V3) for AI-based search and classification (on selected subsets)

# OWI at a Glance — Statistics (Feb 2026)

## Crawling

| | |
|---|---|
| Pages crawled | ~39 B total, ~12 B unique |
| Unique hosts | ~500 M (extrapolated) |
| Crawling effort | 1,927 machine-days |
| Avg. throughput | ~20 M URLs / day |

## Index

| | |
|---|---|
| Documents indexed | ~10.7 B |
| Unique domains | ~75 M |
| Public datasets | 511 main / 1,511 total |
| Index size | 45 TB (main) / 50 TB total |
| Embeddings | 119 M docs · 884 GB |

**Top languages:** English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)

**Multimedia Objects:** Roughly 2-3 quality images and 0.5 Videos per Web-Document

**Comparison:** To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

# Building the Open Web Index

# Conceptual Workflow

## Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.

**(1) Selecting web resources**
Web pages are navigated, prioritized and collected

**(2) Storing web documents**
Multiple gatherers collect web documents and store them in web archives on a European server

**(3) Content extraction**
The content of web documents is extracted (e.g. words, images)

**(4) Metadata extraction**
Metadata (e.g. publisher, author, date) are extracted

**(5) Content analysis**
Features of web documents are extracted (e.g. topic, language, quality, genre, legal constraints, ethical aspects, etc.)

**(6) Index building**
All extracted data from web documents are stored in a specialised database, the so-called webindex

**(7) Index deployment**
The index is deployed in its full version at European data centres or sliced into smaller portions for specific purposes and made available for download

## Search Applications

A user search request will be answered by a search application that makes use of the open web index.

**(a) Selecting web documents**
Web documents are selected that fit to the user search request

**(b) Ranking web documents**
The selected Web documents are sorted (ranked) according to their assumed relevance for the user

**(c) Purpose-specific search**
An application with user interface enables the search for general or specific purposes

**(d) User searches and receives result**
The user is supported to better understand the search process

## Data Products

Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications

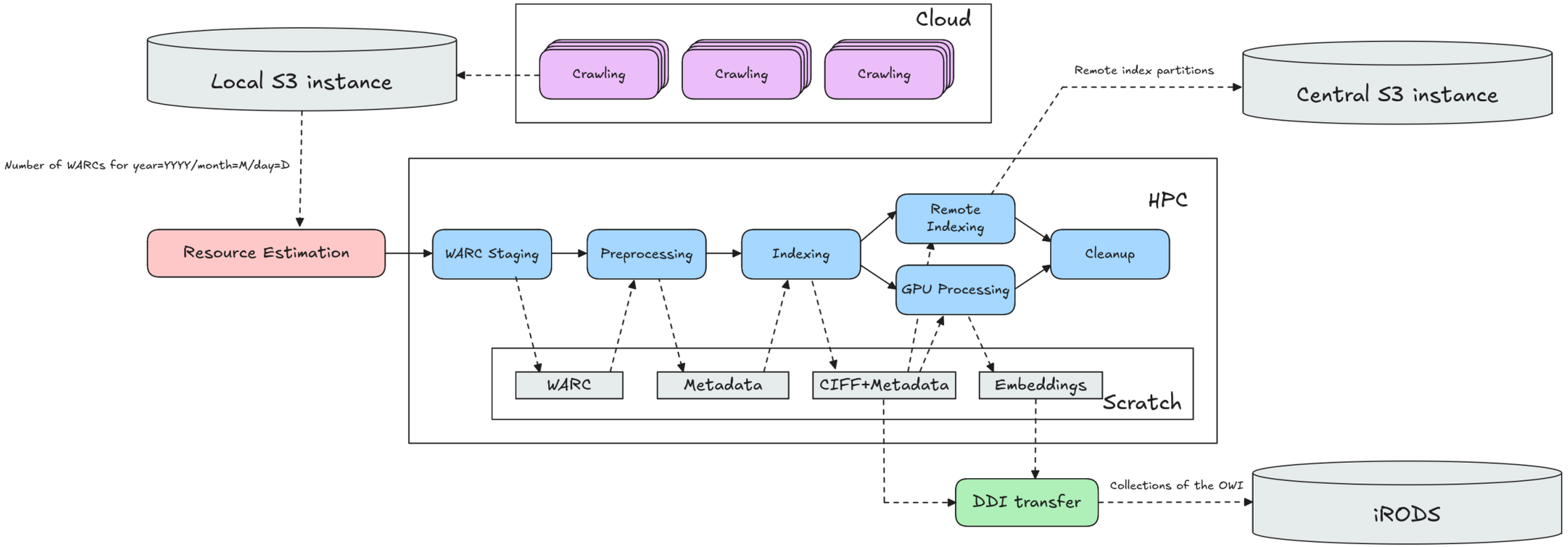
**Building knowledge graphs**
Using the extracted information from web documents, a knowledge graph is created that supports specific search requests

**Building AI Language Models**
Creation of different types of language models by using Web documents

**Any agent, multiple applications**
Language models and knowledge graphs can be used by any agent (or application)

...

# Daily Workflow Pipeline




## Crawlig Infrastructure (2026)

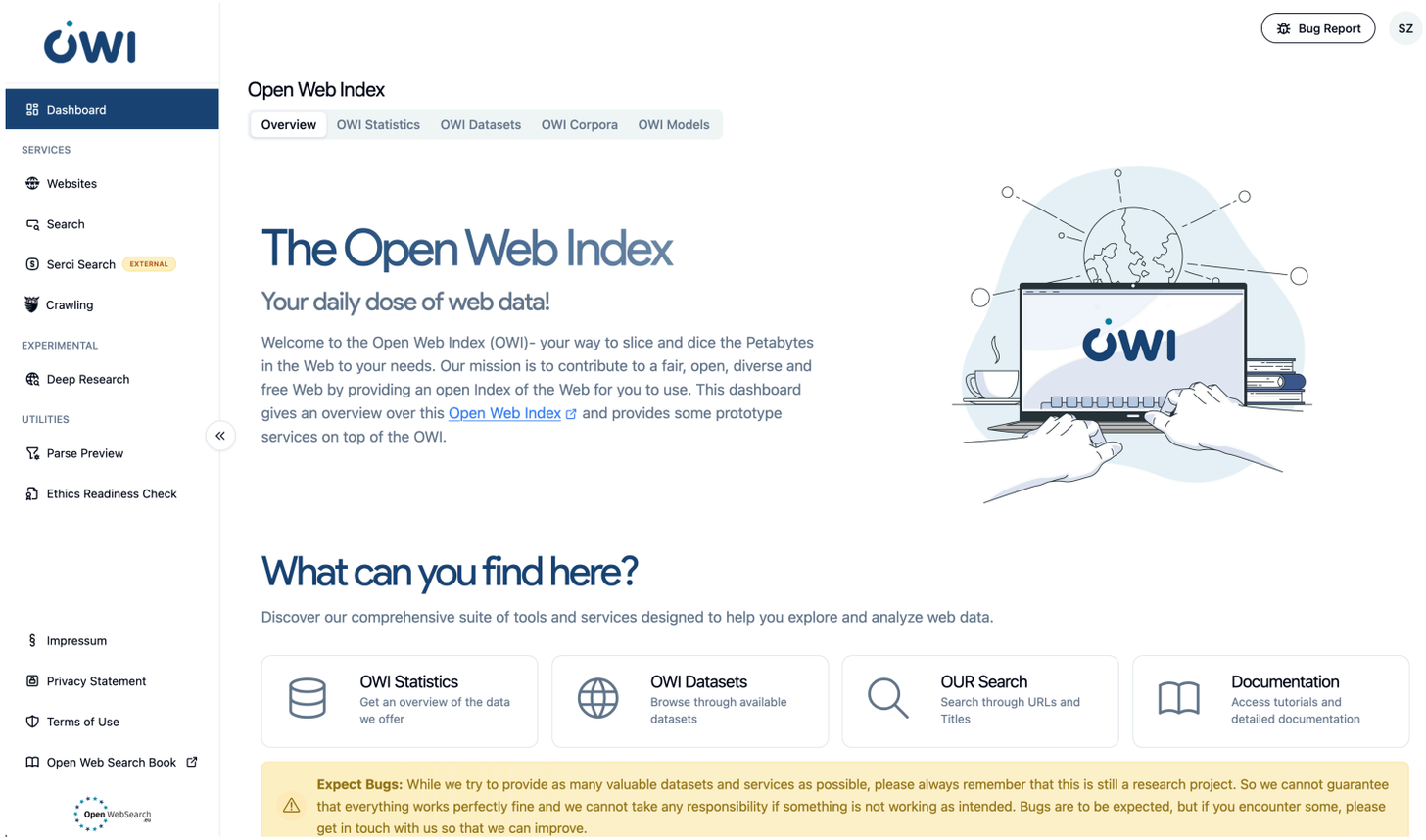
- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

## Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement

# Accessing the Open Web Index

# The Dashboard — openwebindex.eu

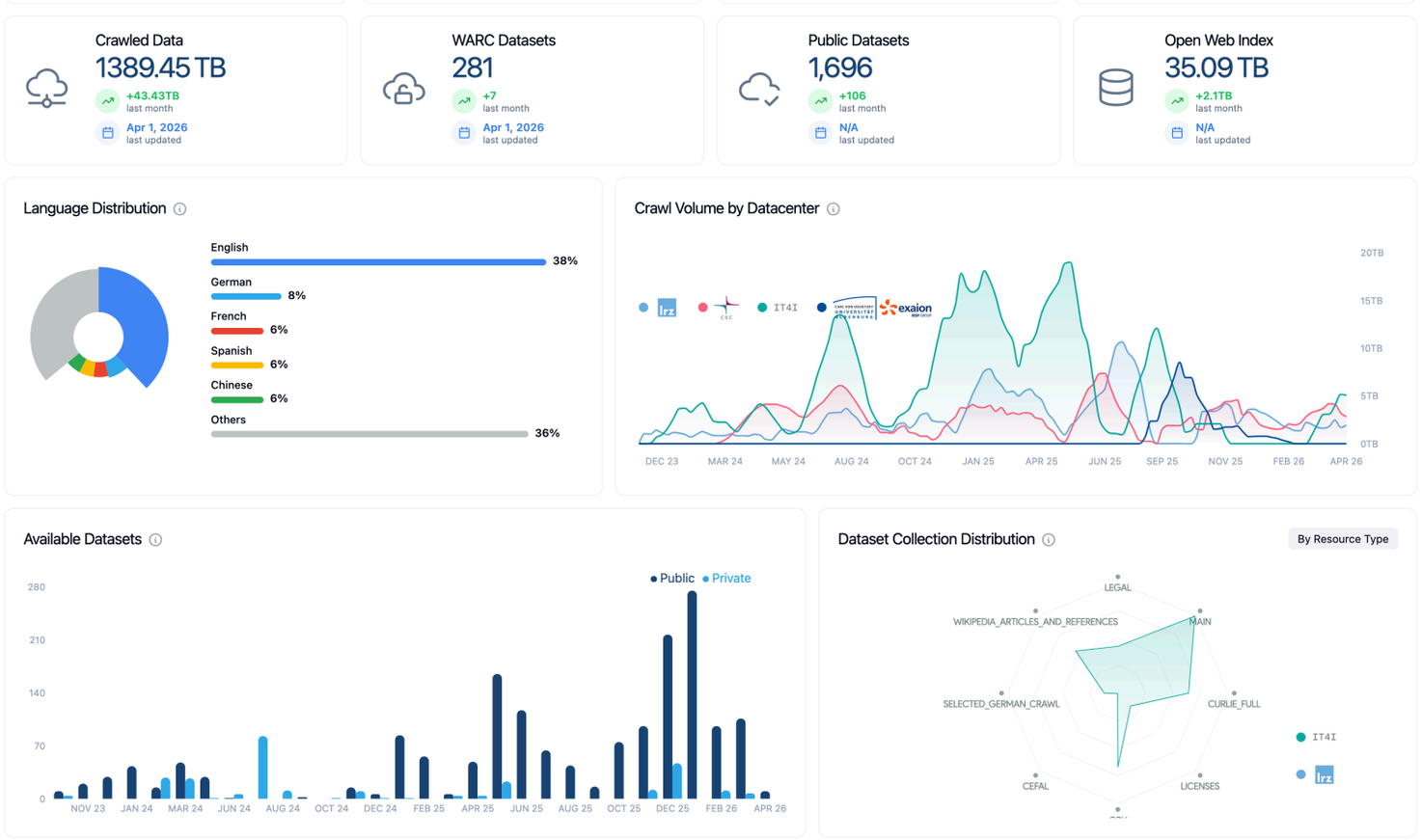



424 registered users · 17,000+ unique visitors · 70,000+ page views

## Central entry point for the Open Web Index openwebindex.eu

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

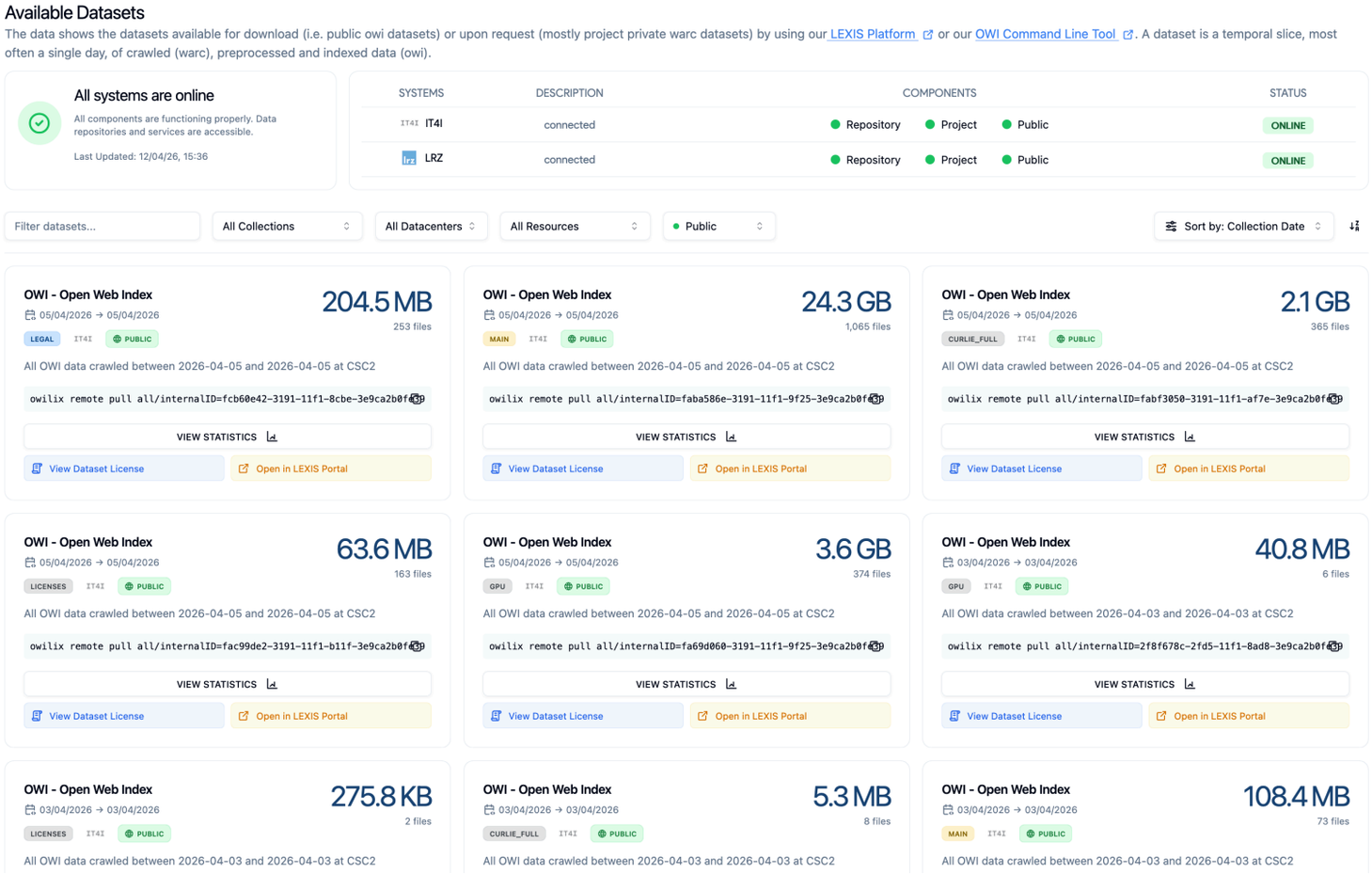
# Dashboard — Statistics & Datasets




## OWI Statistics (04/26)

- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
- Language distribution chart (English 38%, German 8%, French 6%, ...)
- Crawl volume over time per data center + dataset collection distribution by type
- Datasets over time

# Dashboard — Statistics & Datasets




## OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

# Download Tool: owilix

## owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curlie), **sitelist**, **structured data**, **outlinks**

Authentication via B2ACCESS / Keycloak (LEXIS SSO)

### OWILIX - Open Web Index CLI

version 5.3.1 | python 3.11+ | license Apache 2.0

A command-line tool for managing OpenWebIndex datasets across local and remote repositories.

#### Quick Start

##### One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/

### Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

```
[→  ~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
                    Search results (10 results in 113.4s)
```

| score | date | url |
|---|---|---|
| 3.088 | 2025-12-30 | https://easychair.org/smart-program/ICADL2024/person16.html |
| 3.010 | 2026-02-19 | https://www.wolframalpha.com/input/?i=michael |
| 3.010 | 2026-01-19 | https://www.michaelmoennich.com/ |
| 3.010 | 2025-12-08 | https://michaelkoenig.de/ |
| 3.010 | 2026-01-05 | http://www.michaelmoerk.dk/ |
| 3.010 | 2026-01-22 | https://michaelkoenig.de/ |
| 3.010 | 2026-02-18 | https://www.sktthemes.org/forums/users/dedicationpt/replies/ |
| 3.010 | 2026-02-26 | https://www.abeldiaz3.com/tag/michael/ |
| 3.010 | 2026-02-20 | https://discourse.psychopy.org/u/michael |
| 3.010 | 2026-02-27 | https://thedigitalgobag.com/author/michael/ |

## The OpenWebSearch Book

**Comprehensive Documentation at [openwebindex.eu/book](openwebindex.eu/book)**

Covers the full OWI ecosystem:

- Getting started with owilix
- Building vertical search engines with MOSAIC
- Data formats: CIFF, Parquet, embeddings
- Pipeline architecture and HPC deployment
- Legal & ethical guidelines for index users

Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

## Access Modes at a Glance

| Tool | Mode | Best for |
|---|---|---|
| Dashboard | Browser | Explore, monitor |
| owilix | CLI | Download, slice, build |
| Remote index | DuckDB/SQL | Query + download |
| MosaicRAG | Conversational | RAG & chat over OWI |

> All software, data formats and access tools are **open source**

# Using the Open Web Index — Applications

# MosaicRAG — Conversational Search on the OWI

**Modular RAG framework** extending MOSAIC with LLM-based capabilities and dense retrieval (WEBIS / TU Graz)

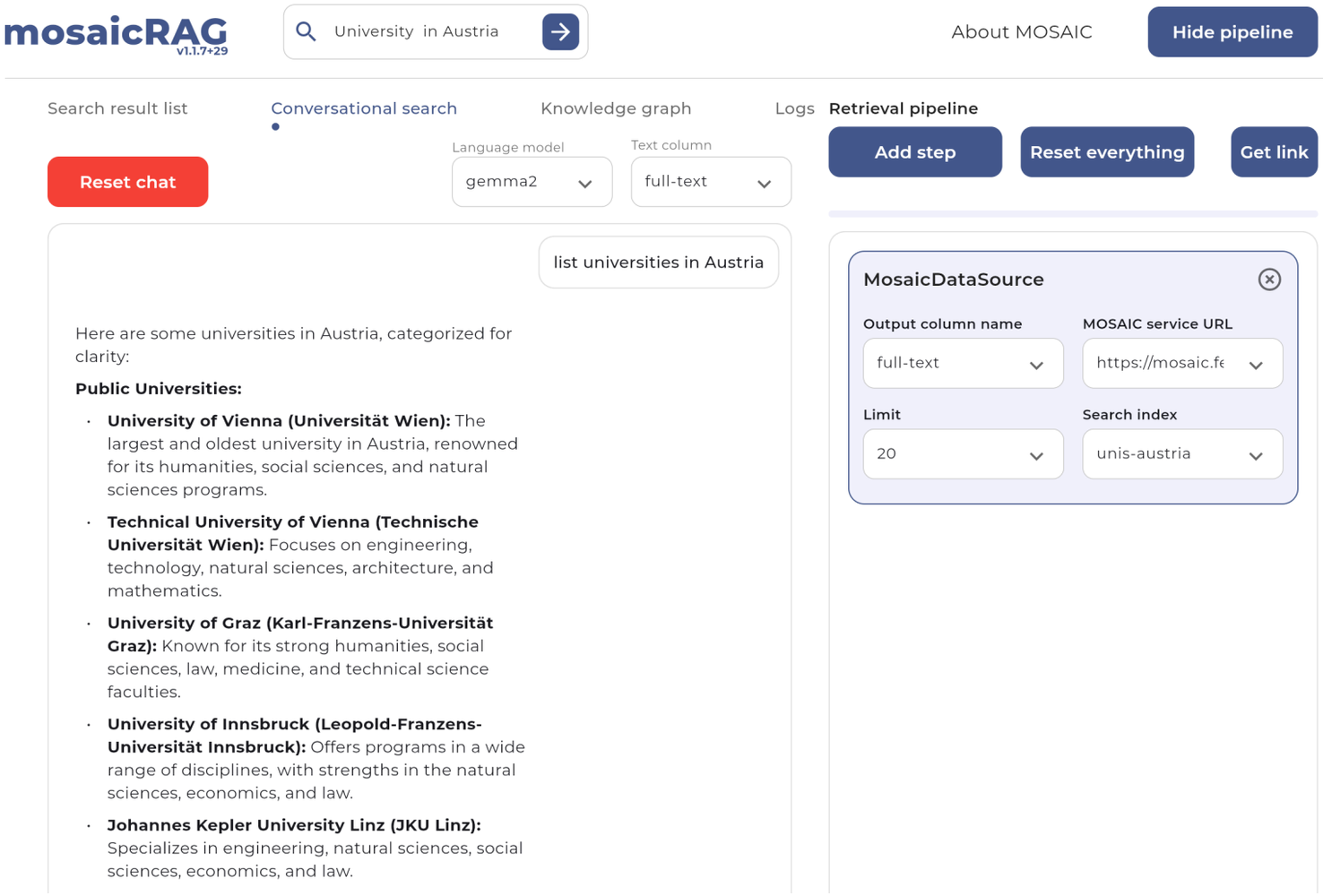
- **Two retrieval paths:** inverted index (CIFF) + dense retrieval
- **4 reranking strategies:** TF-IDF, Embedding cross-encoder, Tournament-style (LLM pairwise), Group-style (LLM window)
- **LLM backends:** self-hosted on-premise models for summarization
- **Conversational search:** follow-up questions about retrieval results; dual-window UI (search results + chat)
- **Workflow:** define retrieval pipeline → issue query → converse about results

### Key Features

- Relevance marking — LLM highlights relevant passages within documents
- Summarization — user-provided instructions applied per document
- Domain-specific indices: science, health, arts (8M+ docs)
- Fully open-source modular pipeline

### Data Pipeline

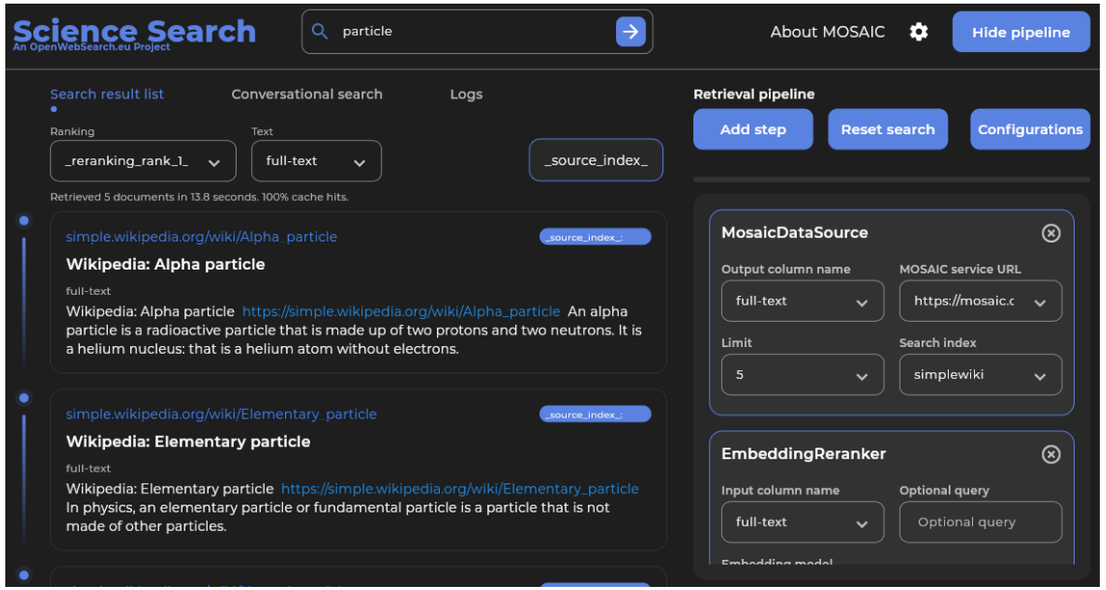
OWI daily datasets → CIFF (inverted index) + Parquet (metadata) + dense embeddings (since Dec 2025) → MosaicRAG




Demo: [mosaicrag.ows.eu](mosaicrag.ows.eu) · [mosaic.ows.eu](mosaic.ows.eu)

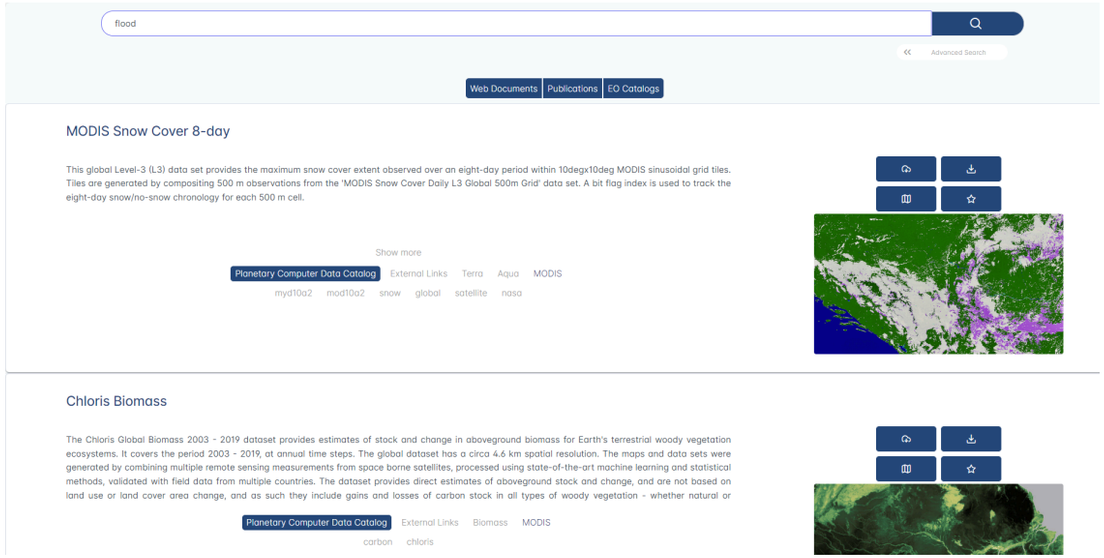
# Mosaic Search Applications · Earth Observation Search

**Mosaic Vertical Search** (TU Graz)
Modular framework for building vertical search engines on OWI slices — domain-specific indices for science, health, arts (8M+ docs)




Demo: [mosaic.ows.eu](mosaic.ows.eu) · [mosaic.ows.eu/demos](mosaic.ows.eu/demos)

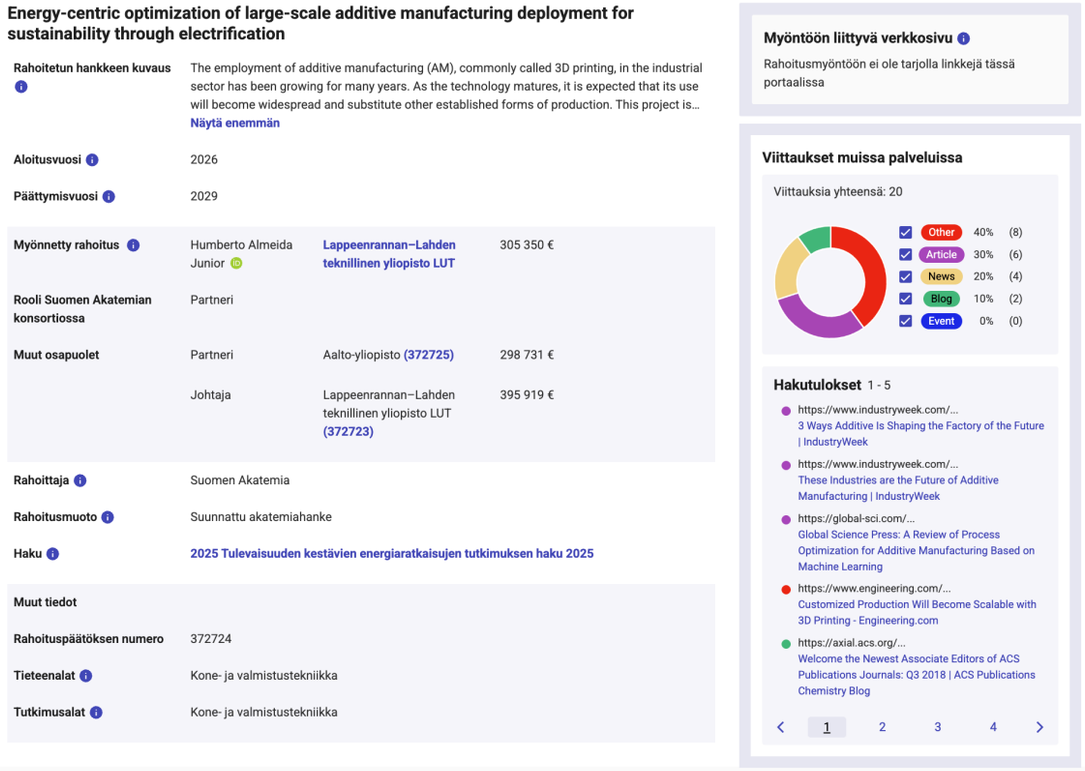
**Earth Observation Search** (DLR / Radboud)
RAG-based QA over scientific papers, datasets & web content; geo-contextualized multi-genre search on interactive maps; KG with 2M+ nodes (OpenAlex, PANGAEA, DLR Geoservice)




Demo: [eo-rag-demo.2.rahtiapp.fi](eo-rag-demo.2.rahtiapp.fi) · [ows-demo.2.rahtiapp.fi](ows-demo.2.rahtiapp.fi)

# Multilingual Scientific Index · Restaurant Recommender

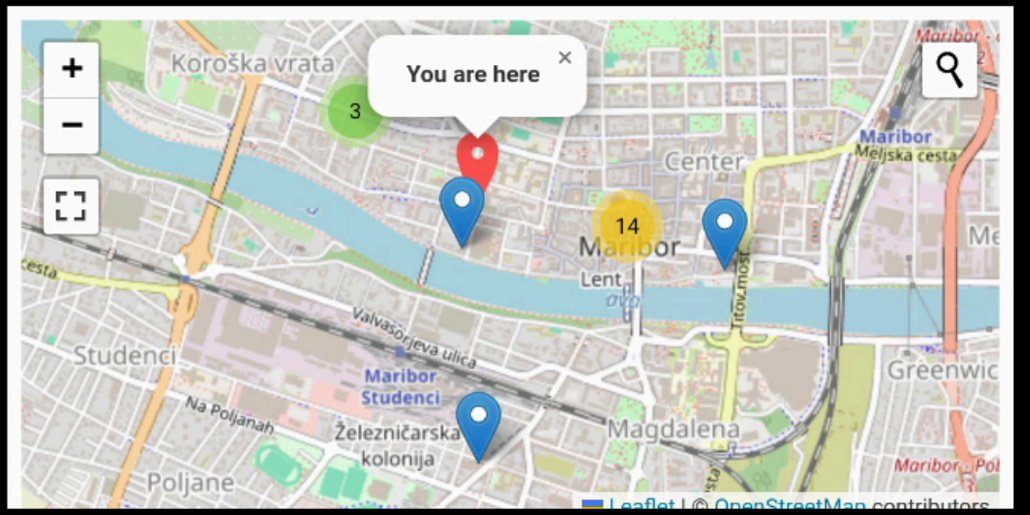
**Multilingual Scientific Index** (CSC)
Enriches Finland's Research.fi portal with OWI web data — Finnish & English science content classified by type (publication, blog, event); embedded directly in national research portal




**Restaurant Recommender** (A1 Slovenia)
Privacy-preserving mobile app; on-device personalization via query imputation; LLM-extracted restaurant features from OWI; 5-month pilot with 214 users — overall satisfaction 4.2 / 5

Demo: research.fi

# Restaurant Recommender




Select City

Or search for restaurants in 1 km radius

Bascarsija grill food

### Baščaršija

Visit Restaurant Page

Poshtna ulica 8, Maribor 2000 Slovenia +386 2 250 63 59

**Cuisines:** Barbecue; European; Grill; Slovenian; Eastern European **Meals:** Lunch; Dinner

**Price:** $4 - $13 **Features:** outdoor seating;

### Okrepcevalnica Sarajevo

Visit Restaurant Page

Gosposvetska cesta 43c, Maribor 2000 Slovenia +386 2 252 34 31

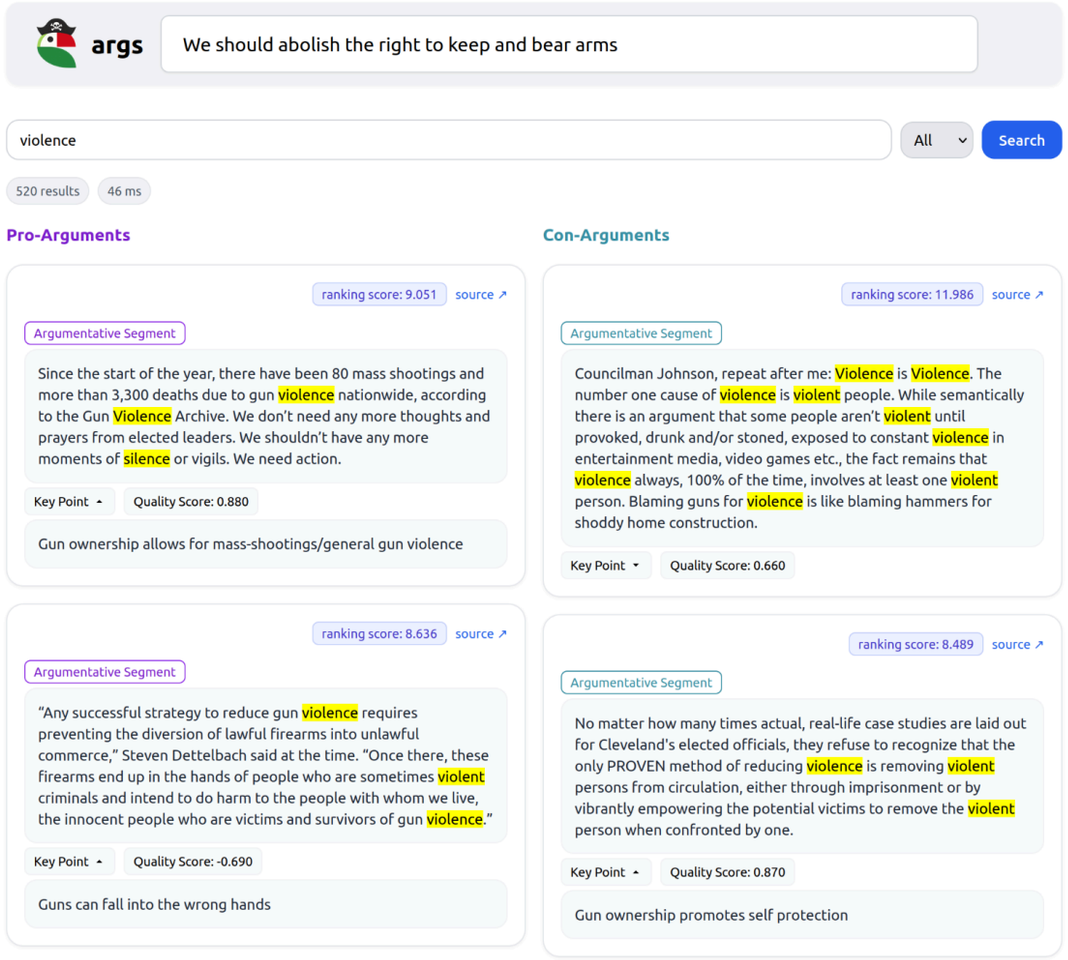
**Cuisines:** Barbecue; European **Meals:** Lunch; Dinner **Price:** $2 - $11 **Features:** outdoor seating;

# Argumentation Search · CERN Search Applications

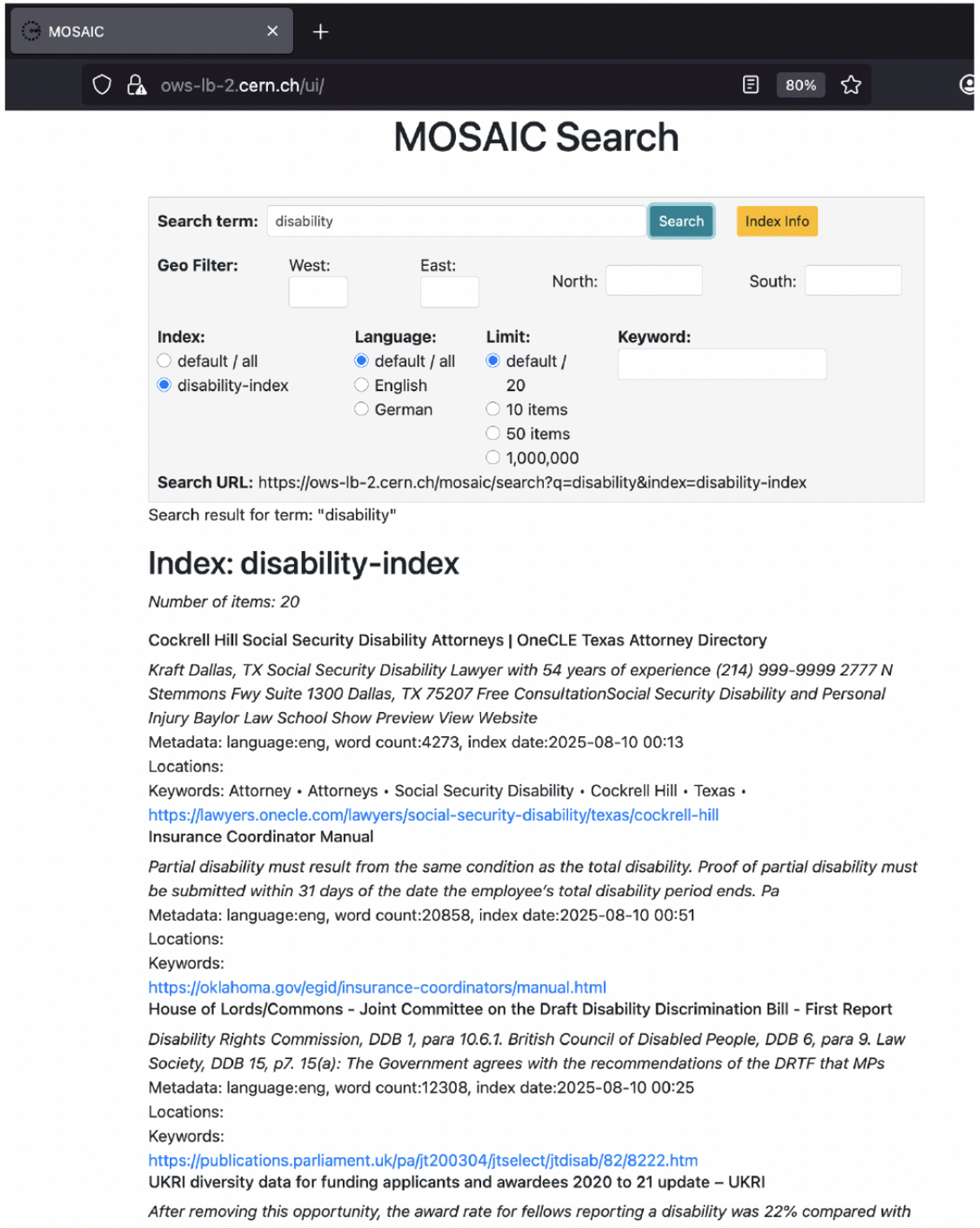
**Argumentation Search** (WEBIS)
args.me extended with OWI: pro & contra arguments on 31 controversial topics; argument quality scoring, key point matching, temporal trend analysis

**CERN Search Applications**

- **Institutional**: 25K+ pages crawled from CERN internal web; full OWS pipeline → MOSAIC index for AccGPT knowledge base
- **Nooon**: disability knowledge search (2.59M docs) for HR & Diversity & Inclusion




Demo: [args.me/owi](args.me/owi) · [args.me](args.me)

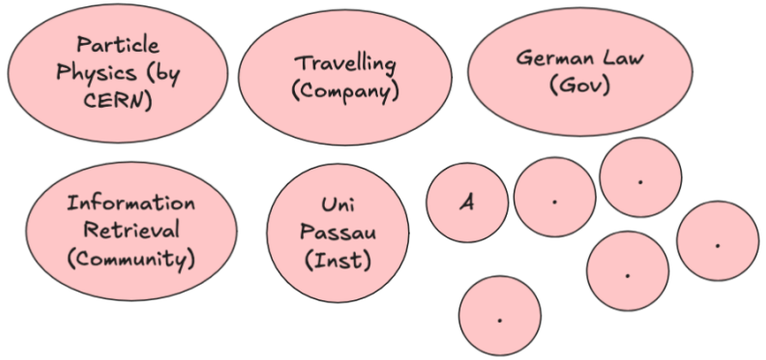



Deployed on [open-science-search.cern.ch](open-science-search.cern.ch)

# Third-Party Projects (FSTPs)

- **AKSE** — argumentation knowledge graph for structured deliberation
- **VERITAS / DEXAI** — verification-oriented RAG for sensitive geopolitical contexts (Ukraine)
- **OMMS** — enriching OpenStreetMap POI with OWI structured data
- **DTCommerce** — product extraction & enrichment pipeline for SME e-commerce
- **CIFFIL** — cross-collection index interoperability for municipal search (Dutch pilots)
- **TILDE** — trustworthy health search with bias detection and entity-level retrieval

# Towards Federated RAG Systems




- RAG / Agentic systems require focused, high-quality collections
- Web information redundancy increases due to LLMs — no one-size-fits-all retrieval
- OpenWebSearch.eu supports low-cost creation of topic-specific RAG systems

**Key ideas:**

- Retrieve from well-designed RAG systems instead of generic search engines
- Topic-specific, deduplicated and curated content
- Domain-optimized ranking, aggregation and summarization
- Lightweight browser-based federation with personal data integration

> **OwlerLite** (WWW 2026) — privacy-respecting browser extension for scope- and freshness-aware retrieval: user-defined **scopes** (trusted sources), semantic freshness detection via SimHash + vector similarity, Scope Fidelity improved from 0.64 → **0.83** (negligible NDCG loss)

# Using the Open Web Index — Data & Analytics

# Using the Open Web Index — Data & Analytics

## Published Datasets

| Dataset | Scale | Use case |
|---|---|---|
| **WebFAQ** | 96M Q&A, 75 langs | QA training, cross-lingual IR |
| **WebFAQ 2.0** | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| **Imprints** | 5.25M hosts, geocoded | Enterprise analytics, Eurostat |
| **German Commons** | 154B tokens | German LLM pre-training |
| **OWS-Curlie-2025** | 20.7M docs + qrels | IR evaluation benchmark |

All datasets available at openwebindex.eu/corpora

## Analytics Potential

- **Official Statistics**: enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring**: supply chain analysis, news monitoring (JRC)
- **AI Training Data**: domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks**: web-scale IR test collections with relevance assessments
- **Trend Analysis**: temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

# Published Datasets

# WebFAQ — Multilingual FAQ from the Web

- **96 million** Q&A pairs across **75+ languages**
- Extracted from structured FAQ markup (JSON-LD, Microdata) in the Open Web Index
- Fields: question, answer, URL, origin, topic, question type
- Parquet format, hosted on Hugging Face

**Use cases:** multilingual QA training, information retrieval, cross-lingual transfer learning

huggingface.co/datasets/PaDaS-Lab/webfaq

| Top languages | Pairs |
|---|---|
| English | ~30M |
| German | ~8M |
| French | ~6M |
| Spanish | ~5M |
| ... | 75+ langs |

# WebFAQ 2.0 — Hard Negatives for Dense Retrieval

- **198 million** FAQ-based Q&A pairs across **108 languages**
- **14.3 million** bilingual aligned QA pairs — largest FAQ-based resource
- **1.25 million** queries with mined hard negatives (20 languages)
- 200 cross-encoder scored negatives per query

**Key contributions:**

1. Novel web crawling for diverse content extraction
2. Hard negatives for dense retriever training
3. Contrastive Learning & Knowledge Distillation
4. Open datasets and training scripts

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*Dinzinger, Caspari, Salman, Topi, Mitrović & Granitzer. "WebFAQ 2.0: A Multilingual QA Dataset with Mined Hard Negatives for Dense Retrieval." arXiv:2602.17327, 2025.*

# Imprints Dataset — Geospatially Grounded Web Curation

- **5.25 million** hosts, **47.8 GB**
- Each website annotated along two dimensions:
  - **Topical classification** via LLM (Foursquare category hierarchy)
  - **Geographic grounding** from legal imprints via geocoding
- **3.08 million** websites geocoded from extracted addresses
- Page types: main page, imprint, contact, about us, privacy, terms

| Country | Geocoded | Share |
|---|---|---|
| Germany | 2,636,913 | 85.5% |
| Austria | 223,805 | 7.3% |
| Switzerland | 160,683 | 5.2% |

| Top-Level Topic | Share |
|---|---|
| Business & Professional | 41.6% |
| Community & Government | 16.2% |
| Arts & Entertainment | 12.2% |

*Dinzinger, Rittlinger, Voigt, Gadziomski, Granitzer, Geiß & Taubenböck. Uni Passau & DLR. License: ODC-BY. openwebindex.eu/corpora*

# German Commons — 154B Tokens for German LLMs

- **154.56 billion tokens**, **35.78 million documents**
- Aggregated from **41 diverse sources** under open licenses
- Quality-filtered, deduplicated, OCR scored

| Domain | Tokens |
|---|---|
| News Commons | 72.67B |
| Cultural Commons | 54.49B |
| Web Commons | 19.89B |
| Political Commons | 3.57B |
| Legal Commons | 2.99B |
| Scientific Commons | 0.84B |
| Economics Commons | 0.11B |

*Gienapp, Schröder, Schweter et al. "The German Commons — 154 Billion Tokens of Openly Licensed Text for German Language Models." arXiv:2510.13996, 2025.*

# OWS-Curlie-2025 — Web Search Test Collection

- **20.7 million** documents from 6 months of OWI crawl data (March–August 2025)
- English documents from the Curlie subcollection (websites labeled in the Curlie directory)
- Three evaluation splits with student-provided topics and relevance assessments
- Jina V3 embeddings, CIFF indexes for sparse retrieval
- Accessible via `owilix` CLI and `ir-datasets-owi` Python package

| Split | Documents | Embeddings | Qrels |
|---|---|---|---|
| `all` | 20,746,181 | -- | -- |
| `radboud-val` | 63,639 | yes | yes |
| `radboud` | 90,852 | yes | -- |
| `jena-kassel` | 77,382 | yes | -- |

*openwebindex.eu/corpora · License: OWIL v1.0*

# Using the Open Web Index — Ideas

# Potential Application Domains

> Some application scenarios discussed at the start of OpenWebSearch.eu are now in reach:

## Web-Analytics for Official Statistics

**Eurostat**: enterprise information extraction from legal imprints — who is doing business where?

**Destatis**: statistical data extraction from web sources to complement national statistics

## Media Monitoring & Intelligence

**JRC Media Monitor**: monitoring media ecosystems and supply chain signals at European scale *(in start phase)*

# A Live Use Case: EU Toy Safety & the Open Web Index

> Can the OWI be used for EU product market surveillance? In preparation of the talk I ran a live exploratory analysis using the OUR$^2$S structured-data API plus some AI Agents.

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*"Make an analysis of the OURS API for the OWI corpus: find children's toys and assess how they appear in the dataset — particularly those potentially violating EU regulations."*

Prompt used in Jetbrain's Junie AI System with Gemini 3.1 Flash

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**The corpus**

| Schema | Documents |
|---|---|
| Web pages | 310 M |
| Structured entities | 137.5 M |
| Unique hosts | 10 M |

**Entities Found**

- **27 M Products** — largest single type
- 26 M Organisations
- 18 M Articles
- 3.8 M Local Businesses
- 2.8 M Events

→ Products are the single largest entity type — directly usable for market analysis.

# A Live Use Case: Products at Scale — Toys in the Index

### Pricing & geography

- ~15 M entities carry price data
- **3 M EUR-priced** products → significant EU market coverage
- Also: USD 2.2 M · UAH 1.9 M · BRL 1.2 M

### Toy query results (live)

- `"children toy"` → **102,530 products**
- With pricing → **51,384** (50 % have structured prices)
- `"Spielzeug Kinder"` → **7,888** German results

### Sample items found

- *"1:24 die-cast metal sports car"* — $90.99
- *"Kids Microscope 1000× Magnifier"* — $59.00
- *"48-teiliges Einhorn Prinzessin Set"* — €8.44

### Structured fields available per product

- Name, brand, GTIN/EAN
- Price, currency, availability
- Description, images, ratings

~50 % of toy products include structured price data, reflecting real e-commerce markup adoption.
~92 % of all entities **lack location data** — a significant data-quality finding.

Whole analysis took 20 minutes in parallel to working on this presentation.

# A Live Use Case: Conclusion

### What the OWI provides today

- Structured Microdata with some features
  - GTIN/EAN presence assessment
  - Brand & manufacturer data
- Completeness and parsing ok, but should be improved

### What would be missing for full EU compliance

- CE marking *(not in schema.org)*
- Age suitability / safety warnings
- Material composition
- ...

Could be added to parser

The OWI is a **foundation for such a use case** — Without an aggregated web index, such an analysis would not be possible. Of course you can use commercial search engines for such tasks, but they are not transparent and you don't know what data they use.

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# Other Research Directions

### Next-Generation Retrieval

- Federated RAG across distributed OWI slices and 3rd-party corpora
- Agentic search orchestration for autonomous knowledge discovery
- Multimodal indexing: images, tables, charts alongside text
- Temporal web search: tracking how the web changes over time

### AI & LLM Infrastructure

- Domain-specific corpora for sovereign European LLM training
- Retrieval-Augmented Generation benchmarks on OWI data
- Open evaluation infrastructure for IR and NLP research
- AI Agents & Search - actively pursued by Uni Passau
- **Multi-loop LLM orchestration**: classify incoming tasks, represent task types as vectors, and use cross-encoder-style scoring to route or sequence specialised LLM loops for the identified tasks
- **Stacked Kohonen embeddings**: map each sentence token to the top-n matching SOM codebook vectors, use their grid positions as explicit embedding coordinates, and condition the next token's SOM lookup on the previous token's match to form a joint sequence vector
- **Capability allocation across parameter space**: which capabilities — knowledge, reasoning, behaviour — should live in *model parameters* vs. *indexed memory* vs. *interaction traces*? Open design question; envisioned

figure maps capability types across the parameter / retrieval / trace space (parked from the TPC26 "Open System Question" slide)

## Societal & Policy Applications

- **Information quality monitoring**: detecting coordinated inauthentic behaviour, SEO spam
- **Digital literacy tooling**: transparent, inspectable search for education
- **Regulatory compliance**: evidence-based auditing of search engine behaviour
- **Digital Sovereignty**: European alternative to US-dominated search infrastructure

The OWI is an open infrastructure — we invite researchers, businesses, and public institutions to build on it

We are looking to be part of interesting research projects to advance the OWI.

We hope to build an open ecosystem that puts open, fair and unbiased information exchange and societal values before commercial interest.

# Conclusion

# In Summary

## What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

## Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

> OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

# Before we come to the Questions

# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

# Get in touch

| | |
|---|---|
| **Website** | openwebsearch.eu |
| **Dashboard** | openwebindex.eu |
| **Code Repository** | https://opencode.it4i.eu/openwebsearcheu-public/ |
| **Community** | openwebsearch.eu/community |
| **Join** | join@openwebsearch.org |
| **General** | community@openwebsearch.eu |

**Thank you — Questions, Comments, Feedback very welcome!**